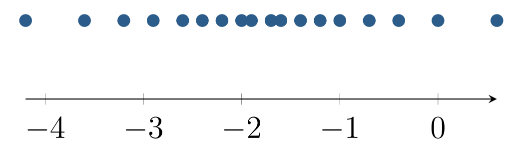
# Two Measurements per Technician

Unit 4 · Topic 4.4 · TODAY'S PROBLEM

Suppose a company takes an SRS of 18 of its 300 technicians. Each completes the same task with old and new interfaces; interface order is randomly assigned. The plot shows $d$ = new time − old time. Researchers ask whether the new interface reduces the population mean time.

**Nia:** "The technician seems to be the independent unit, so I would build one difference per person; the subtraction order matters for the tail."

**Colin:** "There are 36 recorded times, so I would use two independent samples of 18 and a two-sided alternative."




New-interface time minus old-interface time (seconds)

## FIRST TAKE (5 min, on your sheet)

Would you compare each technician's two times or treat them as separate people? Explain in one sentence.

DOK 1 **(a)** Define $\mu_d$ in context.

DOK 2 **(b)** Check the random-sample, 10 percent, and small-sample shape conditions in three short notes.

DOK 3 **(c)** In 3–4 sentences, choose a setup and justify it. State the hypotheses using the given subtraction order and reduction question, then explain why the 36 times are not 36 independent people.

**Optional review:** follow-along (scan the code)

**Work (a)–(c) from the rules on your sheet. Turn in your sheet.**